SMOOTH: An LSL-Middleware Prototype for VR/AR Data Collection

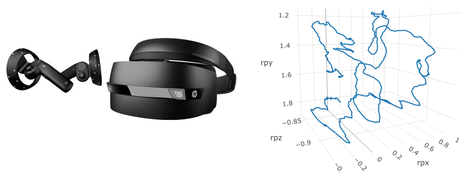
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Introduction

- Virtual reality has great potential for tele-medicine and tele-rehab.
- Without a physical presence, professionals need sensors to get insights from patient (user).
- Reliable data collection and sharing are essential for VR/AR applications

Method

- Windows Mixed Reality(MR) headset data extension to LSL:



(a) (b)

(a) Windows MR headset & hand controllers
(b) data from controller, visualized in RStudio

- Azure Kinect embodiment sensor's images & body tracking extension to LSL:

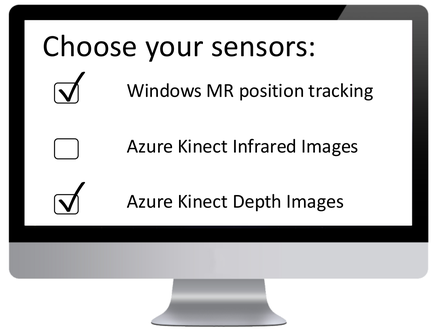


(a) (b) (c)

(a) Azure Kinect sensor, (b) Depth images
(c) Color Images with Skeleton Overlays

Future Work

Design multimodal interface for clinicians & computational biomedicine professionals:



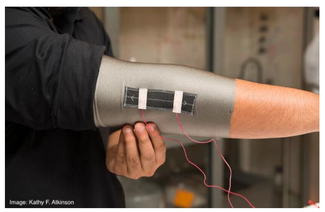
Extend the framework with new VR, smart sensors:



HoloLens 2



Oculus Quest 2



Smart Sleeve

Background

- Lab Streaming Layer (LSL) is able to collect electroencephalogram(EGG) data from many sensors such as:



Tobii Eye Tracker



Wii Remote Motion Sensor



Muse EEG Headgear

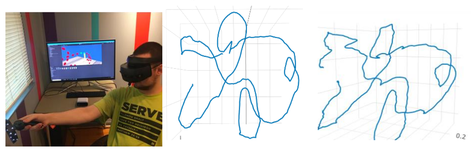
- However: No Virtual Reality and user motion tracking sensors

Results

A pilot tele-rehab study was conducted, using our framework with LSL and Windows MR:

- Users traced a fish shape twice as a therapeutic exercise. N=5 (3 Male, 18.2 $\pm$ 2.8 yrs) healthy individuals.
- Task completion time (TCT):
  - First time: 37.19 $\pm$ 11.89 sec.
  - Second time: 24.61 $\pm$ 4.30 sec.
- A paired t-test showed a significant difference on TCT ($p = .03$) ($t = 3.41$), demonstrating a learning curve for the activity.

(a) tele-rehab pilot activity, and example tracing data from **(b)** Right Hand (dominant), and **(c)** Left hand (non-dominant). b is smoother than c since the user is right-handed.



(a) (b) (c)

Delaware INBRE

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